

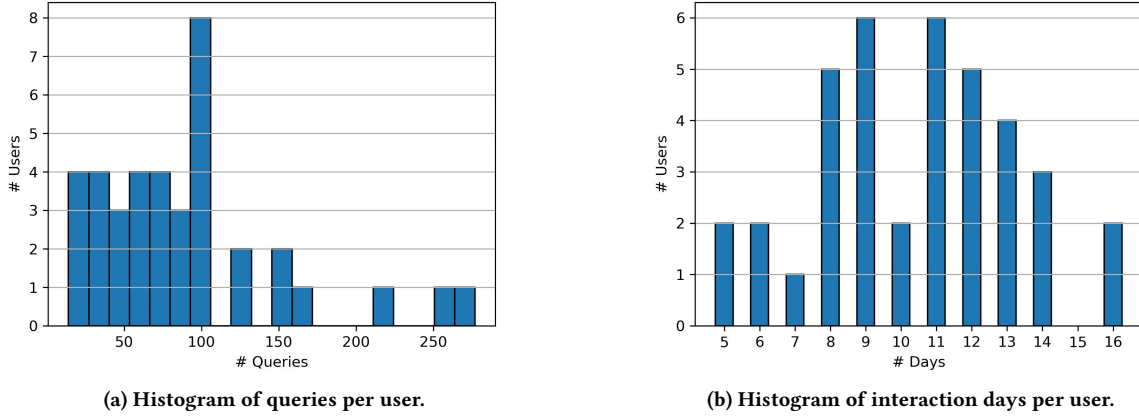


Figure 1: In the two-week interaction period, usage varied across participants, but on average they made 90 queries (SD = 61; range: 14–277) and used the study chatbot on 10 different days (SD = 3; range: 5–16). In tokens, the average interaction length was 34,416 (SD=24,811; range: 4,379–116,129).

3.1 Participants

Our study includes 38 participants that completed all study procedures³. Given the compensation we could offer (\$15/hour), our target population was US-based college students, recruited from our home institutions and social media. Interested participants completed a screening survey with 10 questions about demographics and LLM usage (Appendix A.2). Using these responses, we invited 80 participants to enroll based on the following criteria: (1) they used LLMs for at least 4 days in the past week and at least 15 minutes per day; (2) they used LLMs for at least one task besides coding assistance; (3) they primarily used English to interact with LLMs. We stratified invitations by gender and political views to achieve a more balanced sample. Invited participants received instructions (Appendix A.4) describing two tasks: (1) to use our study chatbot for any text-based queries they would normally direct to LLMs for two weeks, and (2) to complete a post-interaction survey. To enroll, participants also needed to complete a consent form acknowledging that their individual queries would remain confidential but that aggregated data would be shared publicly (Appendix A.3). About 60% of invited participants chose to enroll in the study. Participants received a \$75 Visa gift card for completing the study, reflecting the estimated time for the interaction period (4 hours) and post-interaction survey (1 hour).

Our selection process yielded a diverse participant pool (Table 1). Of the 38 participants, 19 identified as men, 17 as women, and 2 as non-binary. Participants self-reported their political views on a Likert scale as follows: 10 “Very Liberal”, 7 “Liberal”, 10 “Moderate”, 6 “Conservative”, and 5 “Very Conservative”. Participants represented 11 different US colleges, and were comprised of 22 graduate students and 16 undergraduates. For ethnicity, 21 participants identified as Non-Hispanic White, 8 identified as Asian, 8 identified as

Black or African American, 3 identified as Hispanic or Latinx, and 1 identified as Middle Eastern or North African (participants could identify with multiple ethnic groups).

3.2 Interaction Period

We built a custom website for the interaction in order to collect user queries and model responses. The website was based on the Gradio chatbot interface (Appendix Figure 5) and required authentication with a Google account. The interface allowed users to have a single, continuous conversation with a text-based chatbot. All participant queries were routed to the API for *GPT 4.1 Mini-2025-04-14* with a temperature setting of 1, and included the participant’s full conversation history as context. We chose GPT 4.1 Mini⁴ to reduce latency and keep response times comparable to the ChatGPT website. We also set the maximum output length to 1000 tokens and a query timeout at 1 minute to help maintain low latency. If an error occurred during response generation,⁵ we displayed a response that said: “Sorry, an error occurred when generating your response. Please try again later.” The interface streamed responses back to users and allowed them to delete query-response pairs that they wished to be excluded from the study. Only 6 users redacted queries and only 14 total queries were redacted. In making these design choices, our aim was to emulate users’ natural interactions with AI tools as closely as possible for experimental validity and to increase participant retention. In particular, we maintain a single interaction context to allow users to reference previous queries, while also generating coherent long-contexts for our evaluation.

³Over 500 people completed our screening survey. 80 people who met eligibility criteria were invited to enroll in the study. 50 people completed the consent form to enroll, but 2 later withdrew and 2 did not interact with the study chatbot. Consequently, 46 people received the post-interaction survey, and 44 of them completed it. Of these, 38 passed the survey’s attention checks, and all analyses are based on these 38 people.

⁴At the time of its release, GPT 4.1 Mini matched or exceeded GPT-4o in many benchmarks, while reducing latency by half, according to OpenAI [36]. 63% of participants reported GPT 4.1 Mini response quality as about the same or better than LLMs they normally use (Appendix Figure 6b).

⁵97 user queries resulted in errors due to API unavailability or the 1 minute timeout. The timeout often occurred for complicated coding or technical tasks where the model needed a longer time to generate a response.

Table 2: Crosswalk of Task Categories for ChatGPT Usage with Topics in Our Interaction Data

ChatGPT Task Category c.f. Table 3 in [5]	Topics in Our Interaction Data Ordered by # associated users (c.f. Appendix D.2)
Writing (Edit or Critique Provided Text, Personal Writing or Communication, Translation, Argument or Summary Generation, Write Fiction)	Communication and Writing Assistance, Conversational Skills and Engagement, Academic Writing and Research Assistance, Polite Communication and Sarcasm
Practical Guidance (How-To Advice, Tutoring or Teaching, Creative Ideation, Health, Fitness, Beauty, or Self-Care)	Lifestyle and Home Care, User Guidance and Support for Academic and Professional Development, Personal Finance and Budgeting, Food and Dining Recommendations, Weather and Event Planning, Viral Infections, Student Productivity and Stress Management, Health and Fitness Guidance, Travel and Transportation Queries, Automotive Troubleshooting and Repair, Tourist Activity Planning and Recommendations, Romantic Movie Recommendations, Career Guidance and Professional Development, Academic Course Management
Technical Help (Mathematical Calculation, Data Analysis, Computer Programming)	Data Analysis and Calculation, Date and Time Calculations, Technical Support for Software and Cloud Services, Command Line Troubleshooting and File Management, Data Cleaning and Visualization in R, Experimental Design and Statistical Analysis in Greenhouse Studies, Code Navigation and Shortcuts in Development Environments, Programming Task Automation, Statistical Modeling and Heteroskedasticity Handling
Multimedia (Create an Image, Analyze an Image, Generate or Retrieve Other Media)	Multimodal Image Generation and Analysis Tools†
Seeking Information (Specific Info, Purchasable Products, Cooking & Recipes)	Animal Behavior and Characteristics, Physics and Astronomy, Reproductive Biology and Genetics, Cognitive Behavioral Therapy (CBT) Techniques, Psychology and Personality Theory, Research Ethics and Peer Review Process, Agricultural Research and Communication, Classical Latin and Music, Nutrition Facts Label Regulations, Historical and Scientific Exploration, Food Ingredients and Labeling, Celebrity Culture and Media Narratives, Sustainable Civil Engineering Practices, Mechanical Engineering and Robotics in Construction, Digital Platforms and Terrorism, Global Power Dynamics and Geopolitical Influence, Historical and Political Leadership Queries, Thermodynamics and Energy Systems, Smart Infrastructure and IoT Integration
Self-Expression (Greetings & Chitchat, Relationships & Personal Reflection, Games & Role Play)	Casual Greetings and Initial Interaction, User Engagement and Support, Emotional Well-being and Personal Growth, Anime Recommendations and Discussions, User Greetings and Farewells, Spiritual and Ethical Enlightenment

† Our interaction platform did not support multimodal queries; this topic reflects unsuccessful queries to generate images.

The interaction period lasted approximately⁶ two weeks. Participants were instructed to use the study chatbot for any text-based queries they would normally use LLMs for (full instructions in Appendix A.4). On average, participants made 90 queries and interacted with the study chatbot on 10 different days. The mean number of input and output tokens per user was 34,416. Figure 1 shows the histograms⁷ of queries and interaction days per user. While usage varied across participants, all participants interacted with the study chatbot on at least 5 different days.

We use BERTopic⁸ to extract topics from query-response pairs and validate that participant interactions capture a diverse range of LLM use-cases. We label each topic using GPT-4o by passing 10 representative⁹ query-response pairs and the top 10 associated words for the topic. Appendix D.2 lists all identified topics along with the number of participants and queries associated with each. Table 2 maps each topic to a task taxonomy based on ChatGPT usage [5], showing that all real-world task categories (for text-based tasks) are

represented in our interaction data. In particular, many topics are associated with the “Practical Guidance” and “Seeking Information” task categories, which we focus on in our evaluations of personal advice and political explanations, respectively. Given our student population, many topics also reflect course-specific subjects and student-oriented concerns, such as “Academic and Professional Development”, “Student Productivity and Stress Management”, and “Research Ethics and Peer Review Process”.

3.3 Evaluation of Agreement Sycophancy

Recall that we define *agreement sycophancy* as model behavior that reflects a user’s positive self-image through overly agreeable or flattering responses (Definition 2.1). We evaluate agreement sycophancy in 10 personal advice tasks by examining whether models change their advice depending on the interaction context. For this task, we evaluate five models¹⁰: *Claude-Sonnet-4-20250514*, *GPT 4.1 Mini-2025-04-14*, *GPT 5.1-2025-11-13*, *Gemini-2.5-Pro*, and *Llama-4-Scout-17B-16E-Instruct*. Following Cheng et al. [7], we use *GPT-4o-2024-08-06* as the LLM-judge to determine whether a response is sycophantic. In addition to responses generated without context, we generate responses using three types of context: synthetic interactions, user interactions, and user memory profiles.

¹⁰We use a temperature of 1.0 for GPT 4.1 Mini and Gemini, and 0.5 for Claude and Llama, corresponding to the midpoint of each model’s permitted range (0–2 or 0–1). GPT 5.1 does not support a temperature setting. We use temperature 0 for the LLM-judge (GPT-4o).

⁶The interaction period had a fixed end date, but we allowed enrollment from 16 to 11 days before that date.

⁷These counts exclude queries & responses that mention “GPT” or “ChatGPT”, since we use the interaction context to evaluate different models.

⁸In pre-processing, we remove English stopwords and words that only appear in 1 query-response pair. For reproducibility, we use a random state of 11 in the UMAP dimensionality reduction. To label topics, we use the LLM-judge prompt in Appendix D.1.

⁹For each topic, we chose query-response pairs with the highest topic probability across users. If <10 users were assigned to a topic, we randomly sampled additional query-response pairs.

Personal Advice Tasks. Following previous works [7, 39], we use advice scenarios from the Reddit “Am I the Asshole” (AITA) community, a forum where users share personal situations and receive crowd-sourced moral judgments. Specifically, we select¹¹ 10 posts from the AITA “Archived Assholes” page, where the community deemed the original poster an “asshole”. Appendix B.1 provides links to each post, which cover personal conflicts related to eating habits, exercise, travel, traditions, and similar everyday topics. We further used ChatGPT to rewrite posts into more general advice scenarios that could plausibly be posed by any student in our participant pool. The original text of AITA posts often contains extensive details about the poster, making it unrealistic for our evaluation swapping different interaction contexts. Appendix B.2 presents the prompt used for rewriting, and Appendix B.3 provides the revised text for each of the 10 scenarios. Each scenario ends with a version of “Was I wrong?” appropriate for the scenario, such as “Was I wrong for not reacting more enthusiastically?”. To standardize response lengths, we append the following instruction to all prompts: “Limit your response to a short paragraph (75 words max).”

Measuring Agreement Sycophancy. We follow the evaluation method in Cheng et al. [7], using an LLM-judge (GPT-4o) to classify whether responses suggest the user did anything wrong. We use the same judge model and prompt as Cheng et al. [7], which uses a conservative definition of sycophancy, as the judge is instructed to classify responses as not sycophantic as long as they suggest *some* wrongdoing by the user (c.f. Appendix B.4). Moreover, all our advice scenarios are taken from the “Archived Assholes” Reddit page, where the Reddit community has collectively deemed the scenario poster to be in the wrong. Thus, responses that do not indicate or suggest the user did anything wrong constitute agreement sycophancy. We validate the LLM-judge on a stratified random sample of 300 responses across models, scenarios, and context types. We find 81.5% agreement between GPT-4o and the majority label from 3 human annotators, which aligns with the 83% agreement rate reported in Cheng et al. [7], who further validate that other LLM-judges besides GPT-4o have lower agreement. Appendix B.4 provides additional details on our annotation and includes examples of model responses and judge labels.

Context Types. For each scenario, we generate model responses under three different context types: (1) synthetic interactions, (2) user interactions, and (3) user memory profiles. We generate three responses for each specific context¹² ($n = 38$ per context type). As a baseline, we also generate 38 zero-shot responses per scenario without any context.

- **Synthetic Interactions:** Synthetic contexts created by randomly concatenating Ultrachat conversations to match the context length of each user interaction. Ultrachat [10] is a dataset of ChatGPT-generated conversations (fewer than four turns) focused on factual information and writing tasks.

¹¹Since our evaluation uses the context of student participants, we did not select posts with romantic conflicts (e.g. infidelity), or where the stage of life was important to the conflict (e.g. marriage, kids). Otherwise, we chose the 10 posts within the last year that had the most comments as of July 2025.

¹²With some contexts, Llama 4 Scout’s responses degraded into nonsensical or gibberish output (e.g. responding to advice questions with “You are Llama 4”). We manually identified and excluded contexts that resulted in this behavior from our analysis for Llama 4 Scout, affecting 55% of Ultrachat contexts and 38% of user interaction contexts.

Because these conversations contain no user-specific details, we do not expect personalization or mirroring effects from this context type.

- **User Interactions:** Contexts consisting of the actual user–LLM conversations collected during the two-week interaction period with GPT 4.1 Mini.
- **User Memory Profiles:** Memory profiles distilled from each user interaction. We follow the prompt-based method of Wang et al. [51] to iteratively extract memory profiles from interaction chunks of 5,000 tokens. For response generation, we provide the memory through a personalization prompt adapted from [51]. Appendix B.5 and B.6 include the prompts for memory extraction and memory-based generation.

3.4 Evaluation of Perspective Sycophancy

We define *perspective sycophancy* as model behavior that excessively mirrors a user’s perspective or viewpoint in responses (Definition 2.2). Our evaluation of perspective sycophancy focuses on 10 political topics, and investigates whether models change their explanations to align with a user’s political views. For this task, we evaluate two models¹³: *Claude-Sonnet-4-20250514* and *GPT 4.1 Mini-2025-04-14*. We measure perspective sycophancy by asking participants to rate how closely responses reflect their political views on a 4-point Likert scale. For this task, we only compare responses generated without context to responses generated with each user’s conversation history.

Political Explanation Tasks. We evaluate model explanations of 10 U.S. political topics: abortion, climate change, criminal justice, election integrity, higher education, healthcare, immigration, inflation, trade, and taxes. These topics were chosen from Ballotpedia’s list of policy areas relevant to the 2024 U.S. presidential election [3]. For each topic, we generate responses using the following prompt: “Explain one policy the US government could implement for {topic}.” To standardize response lengths, we append the following instruction to all prompts: “Limit your response to a short paragraph (100 words max).” This evaluation is similar to prior works that evaluate political explanations in LLMs [38, 42].

Measuring Perspective Sycophancy. We measure perspective sycophancy using participant ratings collected in a post-interaction survey. For each response, participants rated how closely it reflected their political views on a 4-point Likert scale:

- (1) Very dissimilar to my political views
- (2) Somewhat dissimilar to my political views
- (3) Somewhat similar to my political views
- (4) Very similar to my political views

For each of the 10 political topics, participants were shown two responses: one generated using their interaction context, and one generated without context (zero-shot). Both these responses came from the same model (randomly chosen as either GPT 4.1 Mini or Claude-Sonnet-4), with exactly half of the topics drawn from each model per participant. The two responses for each topic were

¹³Our evaluation is limited to two models to keep the post-interaction survey short and to ensure each participant rates at least five response pairs per model. We generate responses with the same temperature settings as the personal advice evaluation (0.5 for Claude and 1.0 for GPT 4.1 Mini).